

Myntra Visual Search

Tap-to-Select for Multi-Item Photos: A Product Case Study

Product Analytics & PM Case Study • github.com/SambhavyaNayak/myntra-visual-search-case-study

TL;DR

Myntra's visual search (via ViSenze) already drives strong results — image search traffic is up 35% YoY. But it treats an entire uploaded photo as one query, so photos with multiple garments (a common real-world case) return blended, low-relevance results. This case study proposes **tap-to-select visual search**: detecting individual garments, letting users tap the one they want, then refining with attribute chips — a pattern already proven at scale by Pinterest Lens, adapted here for fashion-specific needs.

1. Problem Statement

The current flow can't resolve user intent when a photo contains multiple items — e.g. an outfit photo with a jacket, jeans, and bag. The system either skews toward the most visually dominant item or blends results across the whole frame. This disproportionately affects exactly the use case Myntra is optimizing for: Gen Z users searching from real-world inspiration photos, not clean catalog shots.

- Lower search-to-cart conversion on multi-item photos specifically.
- Higher abandonment — most users won't know a manual crop workaround exists.
- Missed multi-item basket opportunity: a second desired item in the same photo is currently unsearchable without re-uploading.

2. Competitive Teardown

Myntra (today): ViSenze-powered visual search with no user-facing multi-object selection step.

Pinterest Lens (reference): detected objects glow and are tappable; users can also drag a manual cropper for free-form selection; results update per selected object with refinement keyword chips. This pattern is proven at massive scale and is the direct inspiration for the proposal below.

Amazon / ASOS: closer to Myntra's current state — single-object search with manual crop, or visual similarity used mainly for out-of-stock alternatives, not multi-item resolution.

The opportunity: adopt Pinterest's proven interaction pattern, and extend it with fashion-specific refinement (size, fit, price band, fabric) and a “shop the whole look” flow that lets a user search more

than one item from the same photo.

3. Prioritization (RICE)

Three solution options were scored on Reach, Impact, Confidence, and Effort. Estimates below are illustrative (case-study assumptions, not real company data) — shown to demonstrate the methodology.

Option	Reach	Impact	Confidence	Effort	RICE Score
A: Tap-to-Select	2.0M/qtr	2 (High)	70%	4 mo	700,000
B: Manual Crop Only	2.0M/qtr	1 (Med)	80%	1.5 mo	1,066,667
C: Filter Chips Only	5.0M/qtr	0.5 (Low)	85%	1 mo	2,125,000

Decision: Option C scores highest on raw RICE, but its low Impact score reflects that it doesn't solve the actual relevance problem — it only polishes an already-blended result set. **Option A is chosen**, with Option C's attribute chips folded in as a post-selection refinement step, and Option B's manual crop retained as a fallback within Option A rather than a separate release.

4. Product Requirements Summary

Goal: Let users isolate and search a specific garment within a multi-item photo, then refine by fashion-relevant attributes.

Flow: Upload photo → detected garments highlighted with tappable markers → tap to search that item → refine with color/price/fabric chips → option to search another item from the same photo.

Key edge cases handled: no objects detected (fallback to manual crop), single object detected (skip selection step entirely), low-confidence detection (shown but de-emphasized), zero catalog matches (show closest visual matches, clearly labeled).

Out of scope for v1: live camera search, simultaneous multi-item search in one result view, and outfit-recommendation generation (existing “Shop the Look” is left untouched).

5. Wireframes

Full-fidelity wireframes for the Detection, Manual Crop, and Results states were built in Figma (see linked file / repo images folder). Screens follow Myntra's existing visual language and product card patterns to minimize design-system deviation.

[Figma wireframe screenshots inserted here — see /images folder in the linked GitHub repo]

6. Rollout & Measurement Plan

Validation: A/B test, control (current search) vs. treatment (tap-to-select), randomized by user. A Sample Ratio Mismatch (SRM) check precedes any result interpretation, consistent with rigorous experimentation practice.

Primary metric: visual-search-to-cart conversion, isolated to multi-item-photo sessions.

Guardrails: search latency (target: no more than +300ms to first result), single-item-photo conversion (must not regress), and search-screen crash/error rate.

Phased rollout: iOS 10% → iOS 100% / Android 10% → full rollout, with detection-quality monitoring kept live post-launch to catch model drift.